

SymPy Bug Report

1 Bug 58: Left Limit of `elliptic_k(x)` at 1 Returns zoo

1.1 Minimal Reproducer

```
import mpmath as mp
from sympy import elliptic_k, limit, symbols

x = symbols("x")
print("sympy_limit =", limit(elliptic_k(x), x, 1, dir="-"))

mp.mp.dps = 50
print("mpmath_ellipk_1_minus_1e_12 =",
      mp.ellipk(1 - mp.mpf("1e-12")))
```

1.2 Observed Behavior

```
sympy_limit = zoo
mpmath_ellipk_1_minus_1e_12 = 15.201804919087715174172185985894590732575028945707
```

SymPy returns zoo, the unsigned complex infinity, for a real one-sided limit. The nearby high-precision numerical value is finite but already large and positive, matching the expected positive real divergence as the parameter approaches 1 from below.

1.3 Expected Behavior

The mathematically correct result is

$$\lim_{x \rightarrow 1^-} \text{elliptic_k}(x) = +\infty,$$

so SymPy should return ∞ , represented by `oo`, rather than zoo.

1.4 Mathematical Explanation

SymPy's `elliptic_k(m)` is the complete elliptic integral of the first kind,

$$K(m) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - m \sin^2 \theta}}.$$

For real $0 < m < 1$, the integrand is positive. As $m \rightarrow 1^-$, the singularity develops at $\theta = \pi/2$, and the standard endpoint expansion has logarithmic leading behavior

$$K(m) = -\frac{1}{2} \log(1 - m) + O(1).$$

Since $1 - m \rightarrow 0^+$, $\log(1 - m) \rightarrow -\infty$, hence this leading term tends to $+\infty$. The result zoo is therefore not an equivalent representation of the one-sided real limit: it loses the real direction and sign of the divergence.

1.5 Source-Level Diagnosis

The diagnosis locates the relevant implementation in:

`sympy/functions/special/elliptic_integrals.py`

It identifies the `_eval_nseries` method of `elliptic_k` around lines 82–84. The public call enters `Limit.doit` in `sympy/series/limits.py`, proceeds through `gruntz` in `sympy/series/gruntz.py`, and asks for a leading expansion after rewriting the left approach as a small positive dummy. For the representative case, the traced path is

$$\text{elliptic_k}(1 - w) \longrightarrow {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - w\right),$$

where the hypergeometric series routine returns `nan`. The elliptic function’s local series method then also returns `nan`, after which the limit machinery reports zoo.

The problematic rule identified by the diagnosis is that the elliptic integral delegates its local series entirely to a hypergeometric rewrite:

```
return hyperexpand(self.rewrite(hyper)._eval_nseries(x, n=n, logx=logx))
```

For $m = 1 - w$ with $w \rightarrow 0^+$, the needed leading term is logarithmic and real-positive in the limit. The delegated hypergeometric `nseries` path appears not to encode this branch-point expansion, so the one-sided sign information is lost. The exact-value table returning `S.ComplexInfinity` at $m = 1$ is suitable for the singular point itself, but it is not enough to determine the directional real limit. A plausible fix direction, supported by the diagnosis, is to add a branch-aware leading term or series for `elliptic_k` near $m = 1$, at minimum recognizing $m = 1 - w$ with w positive and returning the logarithmic leading behavior.

1.6 Additional Failing Instances

The independent verification checks a shifted family

$$\lim_{x \rightarrow (a+1)^-} \text{elliptic_k}(x - a)$$

for $a = 0, 1, 2, 3, 4, 5$. In every case, the internal parameter $m = x - a$ approaches 1 from the left, so the same logarithmic positive divergence applies. The recorded SymPy behavior is zoo for each shifted instance, while the expected result is ∞ .

Input / Expression	SymPy behavior	Expected behavior
$\lim_{x \rightarrow 1^-} \text{elliptic_k}(x)$	zoo	∞
$\lim_{x \rightarrow 2^-} \text{elliptic_k}(x - 1)$	zoo	∞
$\lim_{x \rightarrow 3^-} \text{elliptic_k}(x - 2)$	zoo	∞
$\lim_{x \rightarrow 4^-} \text{elliptic_k}(x - 3)$	zoo	∞
$\lim_{x \rightarrow 5^-} \text{elliptic_k}(x - 4)$	zoo	∞
$\lim_{x \rightarrow 6^-} \text{elliptic_k}(x - 5)$	zoo	∞

1.7 Related Issues Assessment

The bug-finding harness performed a best-effort deduplication check and classified this as a new instance of a known failure mode. The closest public match reported was SymPy issue #27058, another `elliptic_k` limit problem involving a branch cut approached from imaginary directions at parameter 2. The report also notes issue #26250 and pull request #26264, which concern related elliptic or hypergeometric limit behavior. Those items indicate that the broader elliptic-integral limit and series family is known, but the available evidence did not identify a clear duplicate for the specific real endpoint case $\lim_{x \rightarrow 1^-} \text{elliptic_k}(x)$. This exact reproducible case remains individually actionable and suitable for a targeted regression test.

1.8 Proposed SymPy Regression Test

```
import pytest

from sympy import elliptic_k, limit, oo, symbols

@pytest.mark.parametrize("a", [0, 1, 2, 3, 4, 5])
def test_left_limit_elliptic_k_at_one_is_positive_infinity(a):
    x = symbols("x")
    assert limit(elliptic_k(x - a), x, a + 1, dir="-") == oo
```